Project Prototype
March 30, 2026

Mind the Gap: Graph Neural Networks for NYC Transit Equity
Identifying Underserved Corridors in the MTA Subway Network

Background & Motivation
New York City has one of the most extensive public transit systems in the world, and yet millions of residents live beyond walking distance of any subway station. The MTA's proposed Interborough Express (IBX) serves as acknowledgement that the existing network is insufficient for cross-borough travel/commutes.

{add another paragraph explaining significance}

Recent work has begun applying graph neural networks (GNNs) to urban transit problems. Nagar (2024) demonstrates that GraphSAGE can predict real-time passenger traffic across an existing network, Gamal & Ibrahim (2025) propose a spatio-temporal GNN framework that jointly optimizes route efficiency, energy use, and service accessibility. Both studies, however, operate on networks where service already exists. We build on their approach using real LODES OD flows and a graph-structure ablation to see whether graph neighborhood context improves identification of underserved corridors beyond what edge-level features alone can reveal.

Hypothesis:
"A Graph Neural Network trained on origin-destination commuter demand and subway accessibility data can identify underserved transit corridors, **OD pairs with demand ≥ 6 workers (median demand) where neither endpoint has subway access within 800m (½ mile),** better than a demand-rank heuristic. Its predictions on the Queens-Brooklyn corridor will align with the MTA's proposed Interborough Express, which the model never sees during training."

3 Datasets:

1. LEHD Origin-Destination Employment Statistics (LODES) (Great to determine where people need to go and whether they're currently being served by "accessible" transit)
    a. Description: "The LEHD Origin-Destination Employment Statistics (LODES) data product is released both as part of the OnTheMap application and in raw form as a set of comma separated value (CSV) text files. The LODES data provides detailed spatial distributions of workers and jobs across the United States. It captures home-to-work relationships, employment by industry, worker demographics, and job characteristics at fine geographic levels (the base geographic level in LODES is the census block)."
2. MTA GTFS Stops
    a. Description: "Regular GTFS: This file represents the 'normal' subway schedule and does not include most temporary service changes, though some long term service changes may be included. It is typically updated a few times a year."
3. NYC Census Block Group Shapefile

Approach

We construct a spatial demand graph from two publicly available datasets: the Census Bureau's LEHD Origin-Destination Employment Statistics (LODES), which provides block-level commuter flows, and the MTA's GTFS feed, which provides subway station locations. Census block groups serve as nodes; observed commuter flows serve as edges. Stop count within 800m serves as the primary node feature, encoding transit accessibility.

We will train a GraphSAGE encoder to produce node embeddings that capture each zone's transit context and neighborhood demand structure. An edge classifier then scores each OD pair for the probability of being an underserved corridor, defined as a high-demand connection where neither endpoint has subway access within walking distance.

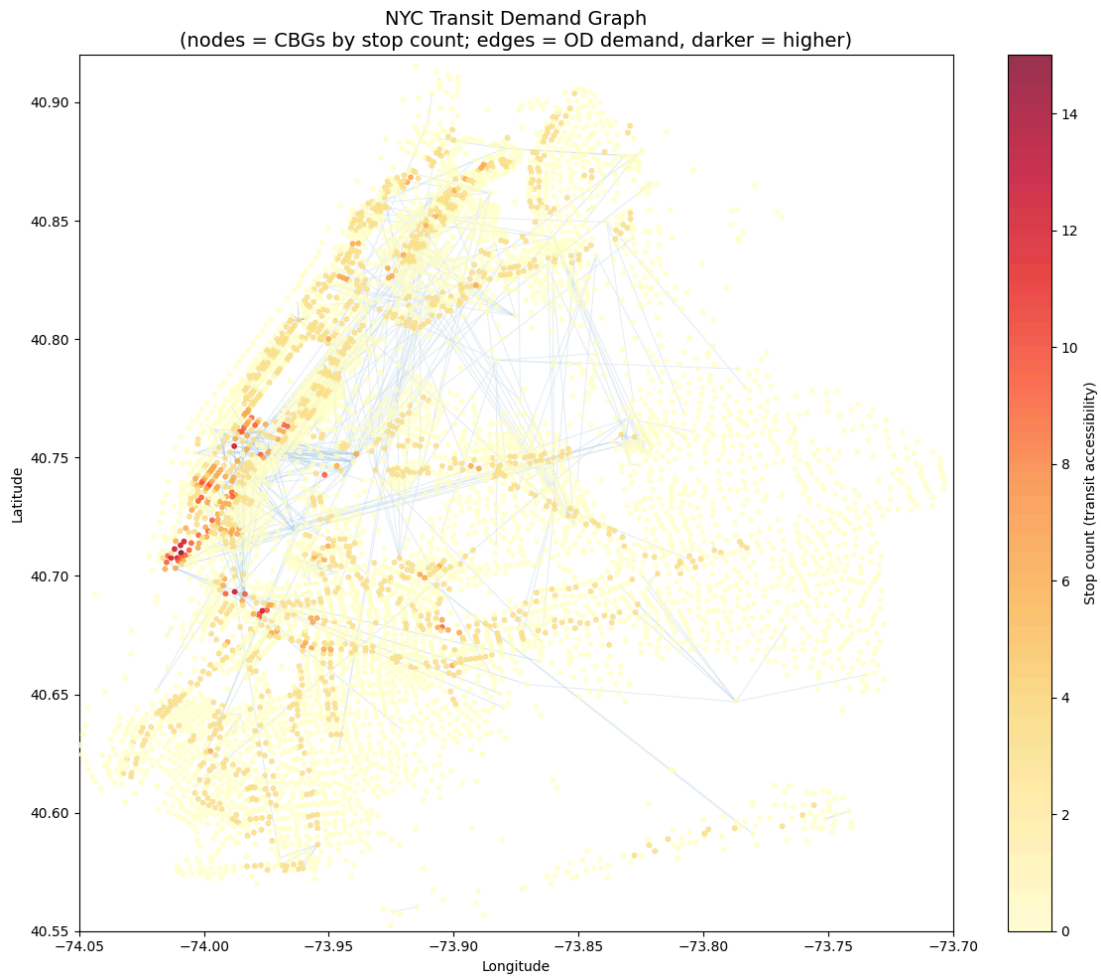
We will compare this approach against a simple demand-rank baseline that flags the top 16.4% of corridors by demand volume as underserved, with no consideration of spatial structure or transit access. Our hypothesis is that the GNN's ability to aggregate neighborhood-level information, seeing not just an individual corridor's demand, but the broader pattern of demand and access surrounding both endpoints, will produce meaningfully better identification of underserved corridors as measured by F1 score on held-out (testing) data. We use F1 as our metric because a high F1 requires both a high precision and high recall.

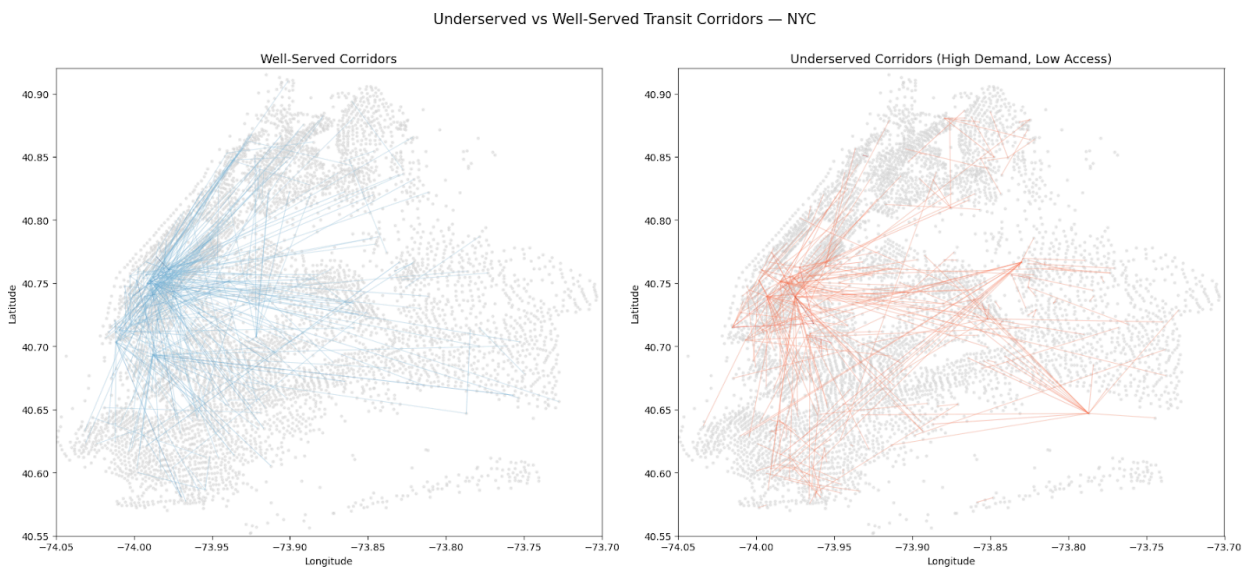
Precision: of all the corridors the model flags as underserved, what fraction actually are?
Recall: of all the truly underserved corridors, what fraction did the model catch?

The GNN, through message passing, learns that a high-demand corridor between two zones that are both surrounded by other low-access zones is structurally different from a high-demand corridor in a dense, well-served neighborhood. That spatial context is invisible to the baseline but learnable by the GNN.

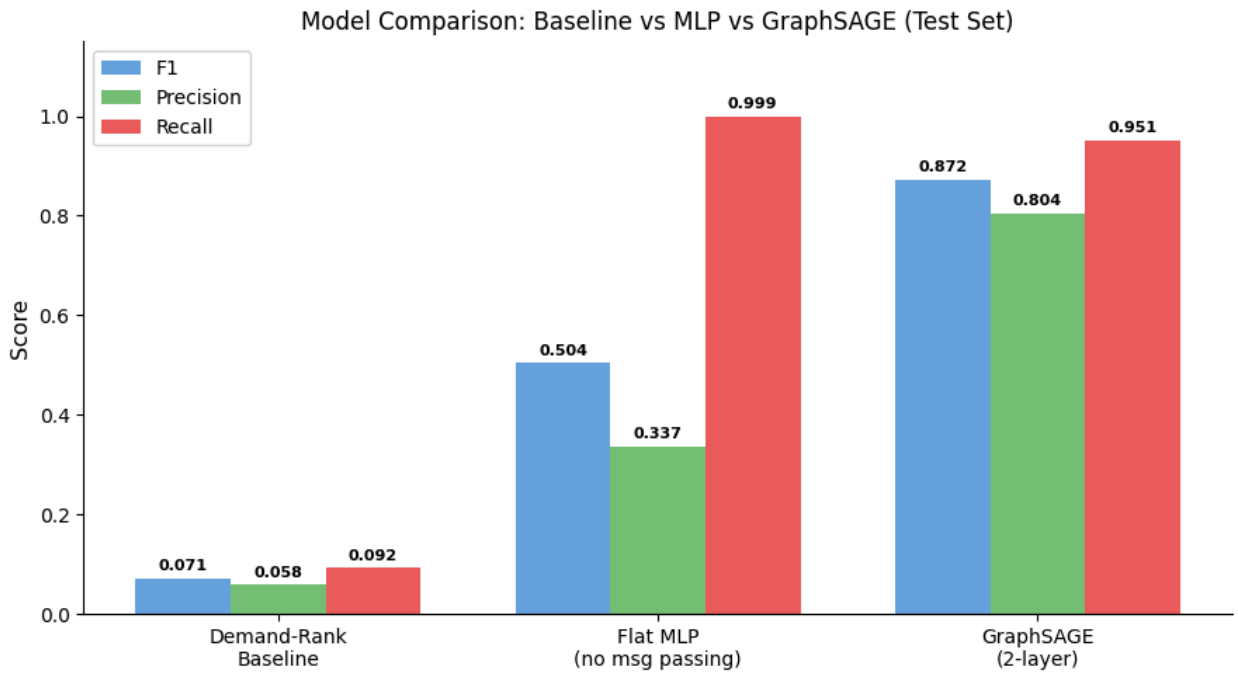
We will examine the Queens-Brooklyn IBX corridor to evaluate whether the model's predictions align with expert planning judgment and to characterize what the model gets wrong.

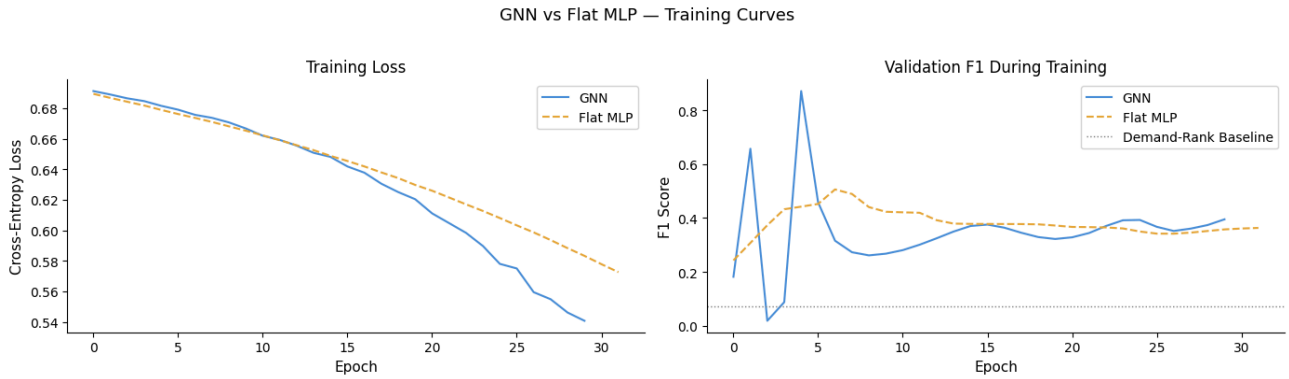
Graphs produced from sampled data

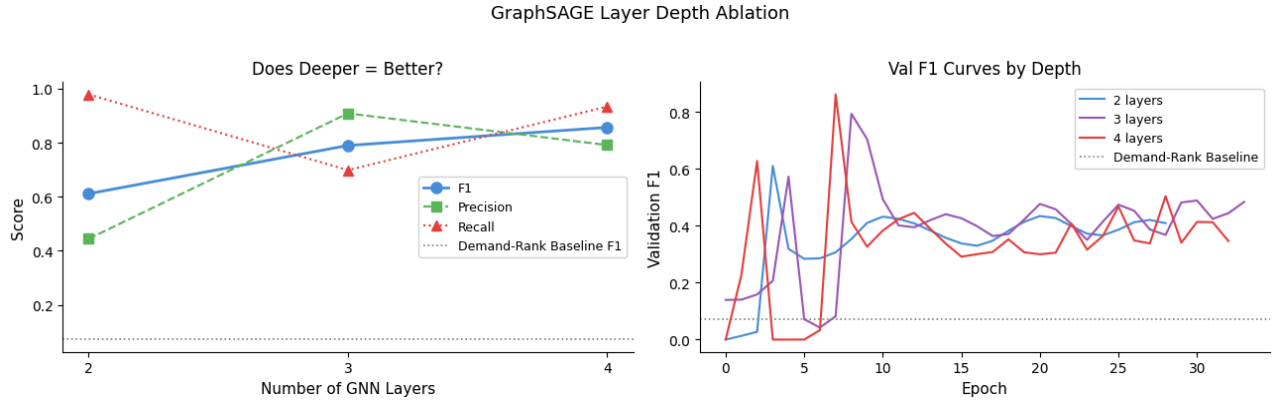




Model assessment figures


Model Comparison: Baseline vs MLP vs GraphSAGE (Test Set)
F1
Precision
Recall
Score
0.0
0.2
0.4
0.6
0.8
1.0
0.071
0.058
0.092
0.504
0.337
0.999
0.872
0.804
0.951
Demand-Rank Baseline
Flat MLP (no msg passing)
GraphSAGE (2-layer)


GNN vs Flat MLP — Training Curves
Training Loss
Cross-Entropy Loss
Epoch
GNN
Flat MLP
Validation F1 During Training
F1 Score
Epoch
GNN
Flat MLP
Demand-Rank Baseline


GraphSAGE Layer Depth Ablation
Does Deeper = Better?
Score
Number of GNN Layers
F1
Precision
Recall
Demand-Rank Baseline F1
Val F1 Curves by Depth
Validation F1
Epoch
2 layers
3 layers
4 layers
Demand-Rank Baseline